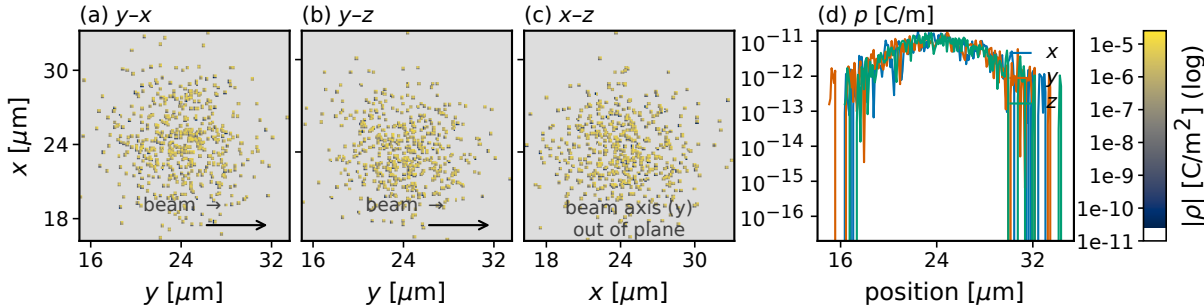
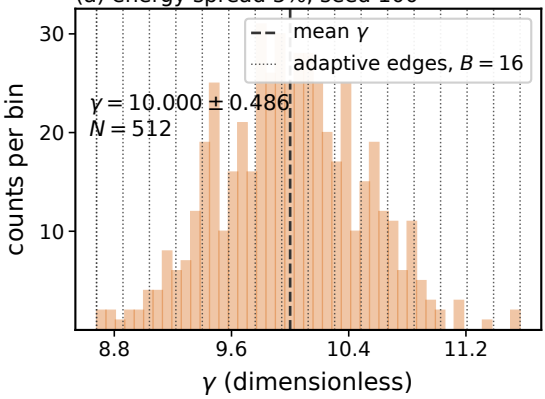
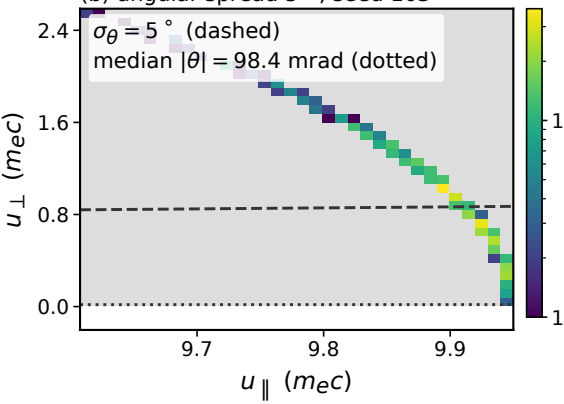
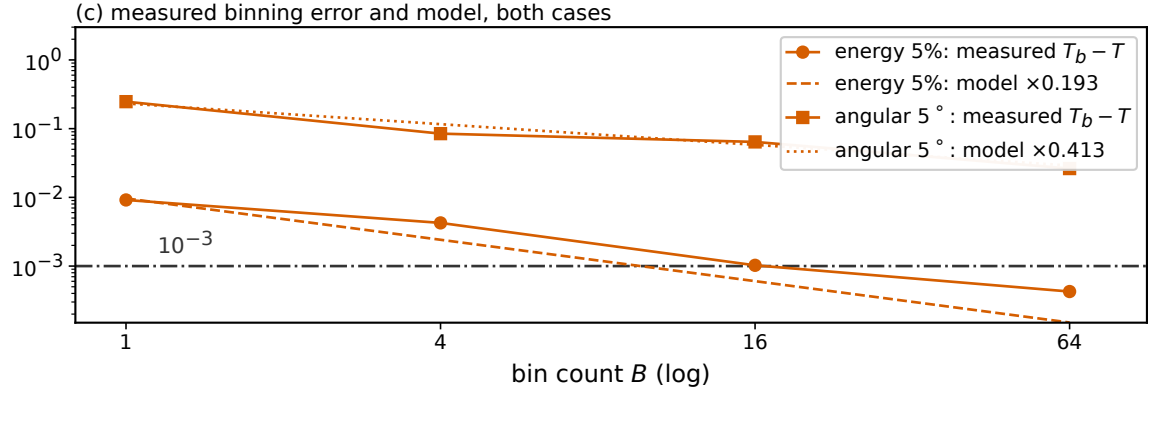

(a) y–x
(b) y–z
(c) x–z
(d) p [C/m]
x [μm]
y [μm]
y [μm]
x [μm]
position [μm]
beam →
beam →
beam axis (y)
out of plane
x
y
z
|ρ| [C/m²] (log)


(a) energy spread 5%, seed 100
mean γ
adaptive edges, B = 16
γ = 10.000 ± 0.486
N = 512
counts per bin
γ (dimensionless)


(b) angular spread 5 ° , seed 103
σθ = 5 ° (dashed)
median |θ| = 98.4 mrad (dotted)
u⊥ (mec)
u∥ (mec)


(c) measured binning error and model, both cases
energy 5%: measured Tb − T
energy 5%: model ×0.193
angular 5 ° : measured Tb − T
angular 5 ° : model ×0.413
10−3
bin count B (log)
Tb − T, peak-normalised, 8 ≤ d ≤ 16 cells (log)